

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-2

Faculty

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Electrical Energy Sources



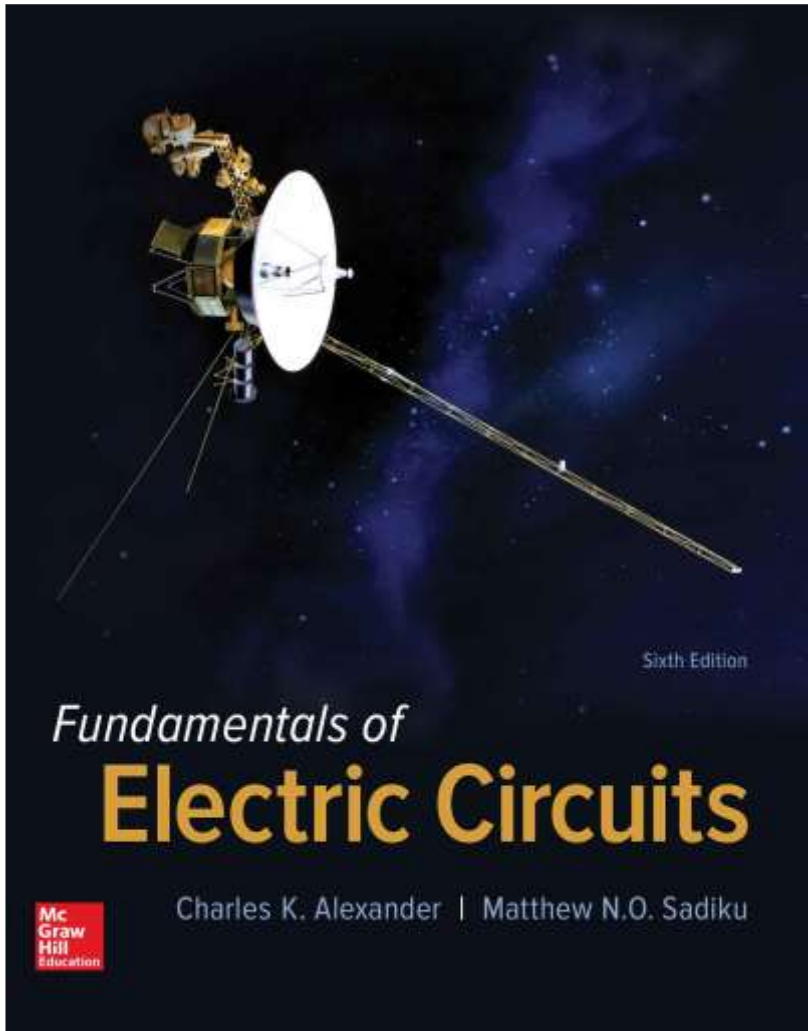
What's the difference?

Syllabus

Course Code:ECE1002	Course Title: Fundamentals of Electrical and Electronics Engineering	TPC	3	2	4
Version No.	1.2				
Course Pre-requisites/ Co-requisites/	None				
anti-requisites (if any).	ECE1005				
Objectives:	1. Equip student with fundamental concepts of different circuit elements. 2. Develop understanding of various circuit laws, theorems and analysing circuits. 3. Help student to carry out projects which contain electrical and/or electronics modules.				
Expected Outcome:	On completion of the course, students get the ability of 1. Understand the characteristics of different electrical and electronic components and circuits 2. Solve DC & AC Circuits 3. Understand the working of electromechanical machines 4. Apply this knowledge in interdisciplinary projects				

Module No. 1	DC Circuit Analysis	10 Hours
DC Circuit Basics: Basic terminology recap (Conductors, Semiconductors, Insulators, Electric charge, Current, Resistance, Potential, Potential difference, Voltage, Work, Power, Energy – with units, Electric circuit, mesh, loop, node). Classification of circuit components (active, passive), Ohm's law, Kirchoff's law, nodal and mesh analysis, source transformation, star, and delta transformations. Cost of energy calculations. Sources (Voltage, Current – dependent and independent), R, L, C, and their calculations Circuit Theorems: Thevenin's theorem, Superposition theorem. Maximum power transfer theorem.		
Module No. 2	AC Circuit Analysis	10 Hours
Single-phase systems: Introduction, AC terminology, values (RMS Values, Average value, Form factor) of different signals. Concept of j-operator, impedance, and admittance. Power in AC circuits, problems related to calculation of active, reactive, and apparent power. Concept of power factor and its significance. Three-phase systems: Introduction, Balanced systems, comparison between star and delta connection of loads. Calculation of three-phase voltage, current, power. Two wattmeter method for measuring three-phase power with examples.		

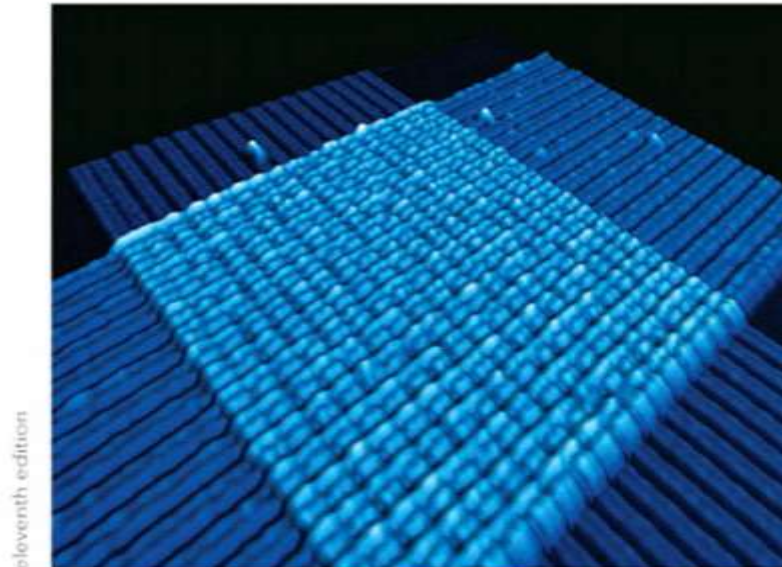
Module No. 3	Semiconductor Devices and Applications	10 Hours
Conduction in Semiconductor materials; PN junction diodes: IV characteristics and current equation, Rectifier circuits design, and analysis; Zener diode and applications, BJT: BJT characteristics and biasing, BJT as an Amplifier and switch; Introduction to MOSFET;		
Module No. 4	Operation Amplifier (Op-amps)	8 Hours
Ideal Op-amp, Differential amplifier: differential and common mode operation, common-mode rejection ratio (CMRR), Practical op-amp circuits: inverting amplifier, non-inverting amplifier, weighted, summer, integrator, differentiator, Large signal operation of op-amps;		
Module No. 5	Electrical Machines	7 Hours
DC Machines: Analogy between electrical & magnetic circuits, Types of electrical machines, construction, working principle, and laws. EMF equation of generator, Torque equation of motor.		
AC Machines: Single phase transformers – working principle, EMF equation, Voltage transformation ratio, types, and construction; Different loss in transformer		



- Module-1 :DC Circuits Analysis
- Module-2 :AC Circuits Analysis

electronic devices and circuit theory

ROBERT L. BOYLESTAD | LOUIS NASHESKY



- Module-3 :Semiconductor Devices and Applications
- Module-4 :Operational Amplifiers

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ELECTRIC MACHINES

FOURTH EDITION

D P Kothari

*Vice Chancellor
Vellore Institute of Technology (VIT)
Vellore, Tamil Nadu*

I J Nagrath

*Adjunct Professor
Birla Institute of Technology and Science (BITS)
Pilani, Rajasthan*



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- Module-5 :
- Electrical Machines

Text Books

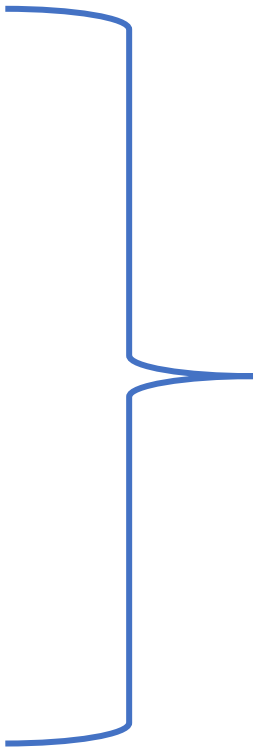
1. Charles K Alexander, Mathew N O Sadiku (2012), Fundamentals of Electric Circuits, 5th edition, Tata McGraw Hill.
2. S K Bhattacharya, Basic electrical and electronics engineering, 2nd edition, 2018, Pearson.
3. William H. Hayyt, E. Kemmerly and Steven M. Durbin “Engineering circuit analysis” 8th edition, McGraw Hill 2013

References

1. S. B. LalSeksena, Kaustuv Dasgupta, Fundamentals of Electrical Engineering, Cambridge University Press, 2016.
2. John Bird (2010), Electrical circuit theory and technology, 4th edition, Newnes publications.
3. Fitzgerald, Higgabogan, Grabel (2009), Basic Electrical Engineering, 5th edition, McGraw Hill.

Evaluation Methods:

- CAT-1
- CAT-2
- FAT
- Quizzes
- Assignments
- LAB

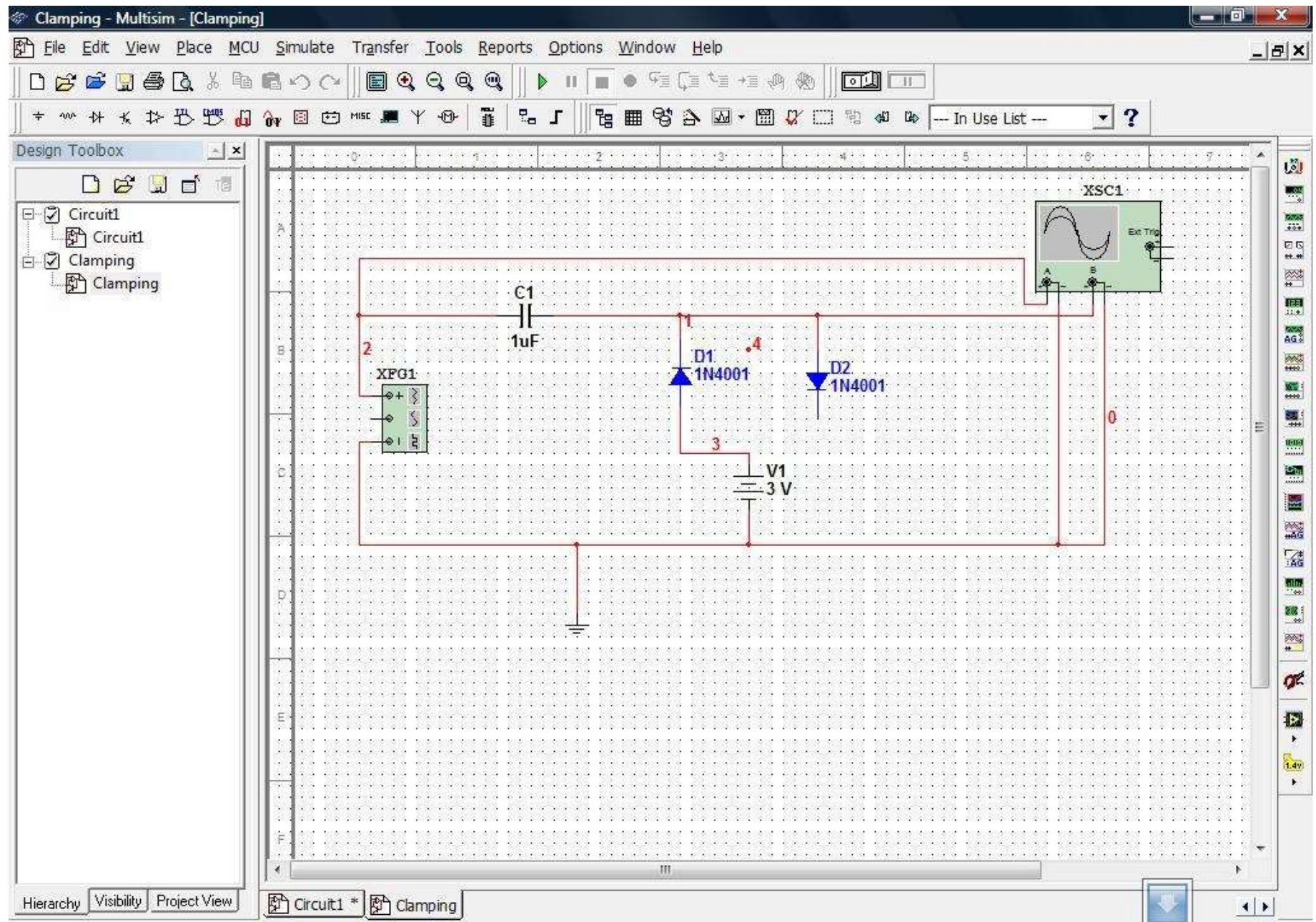


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LAB



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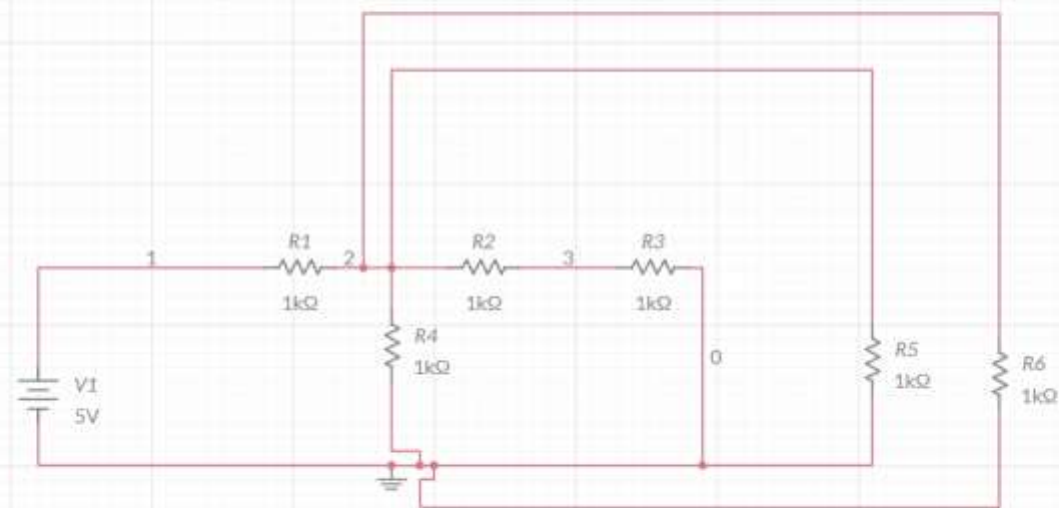
Schematic



Grapher



Split



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- <https://www.coursera.org/courses>
- <https://nptel.ac.in/course.php>
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